

Systems of Particles and Rotational Motion

Particle - It is ideally represented as a point mass having no size.

Rigid Body

→ It is a body with a perfectly definite and.

e.g. Rod, cylinder, sphere, disc etc.

* In real, no body is perfectly rigid.

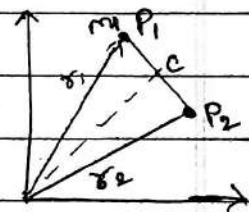
Centre of Mass

- The centre of mass of a body is a point where the whole mass of the body is supposed to be concentrated.

- Centre of gravity = whole weight is concentrated at point.

- COM of a two particle system:

$$\left[\vec{R}_{\text{com}} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} \right] \text{ [It is dis. From any origin]}$$



- Centre of mass for continuous bodies

$$\left[\vec{R}_{\text{cm}} = \frac{1}{M} \int \vec{r} dm \right]$$

- * COM of a triangle lies on point of concurrence of the medians i.e. on centroid.

COM of some regular bodies

Shape of body	Position of COM
1. Long thin rod	middle point of Rod
2. Thin circular ring	Geometrical centre of ring.
3. circular disc.	" " disc
4. Rectangular lamina	Point of intersection of diagonals
5. Rectangular cubical block	" "
6. cylinder	Middle point of Axis
7. Solid or hollow sphere	Geometrical centre
8. Triangular lamina	Centroid
9. Right circular cone	At point on its axis at distance of $h/4$ from base.

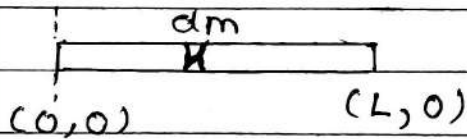
COM of Continuous Bodies

$$x_{com} = \frac{\int x dm}{\int dm}$$

$$y_{com} = \frac{\int y dm}{\int dm}$$

1. Rod (1-D)

(1-D) $\rightarrow$ Mass per unit length = λ



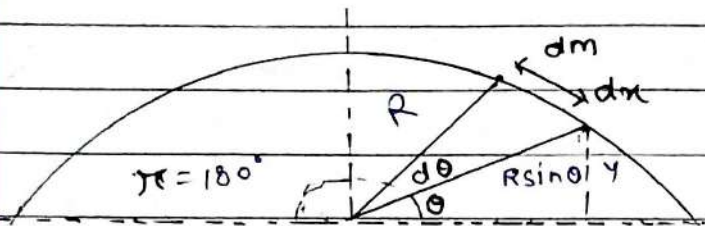
$$\lambda = \frac{dm}{dx}$$

$$\lambda = \frac{M}{L}$$

$$dm = \lambda dx$$

$$x_{com} = \frac{\int x dm}{\int dm} = \frac{\int x \lambda dx}{\int dm} = \frac{\lambda}{M} \left[\frac{x^2}{2} \right]_0^L = \frac{\lambda}{\lambda M} \times \frac{L^2}{2} = \frac{L}{2}$$

2. COM of semicircular wire (1-D)



$$\theta = \frac{\text{Arc}}{\text{Radius}} \Rightarrow d\theta = \frac{dx}{R}$$

$$\Rightarrow dx = R d\theta$$

$$\lambda = \text{mass per unit length} \quad y_{\text{COM}} = \int y \frac{dm}{dm}$$

$$\lambda = \frac{dm}{dx}$$

$$\frac{dm}{dx} = \lambda \quad \left[\lambda = \frac{M}{\pi R} \right] \quad = \frac{\int_0^\pi R \sin \theta \lambda R d\theta}{M}$$

$$= R \lambda = \int_0^\pi \sin \theta \cdot d\theta$$

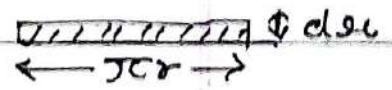
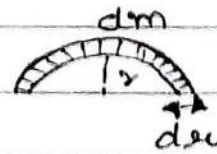
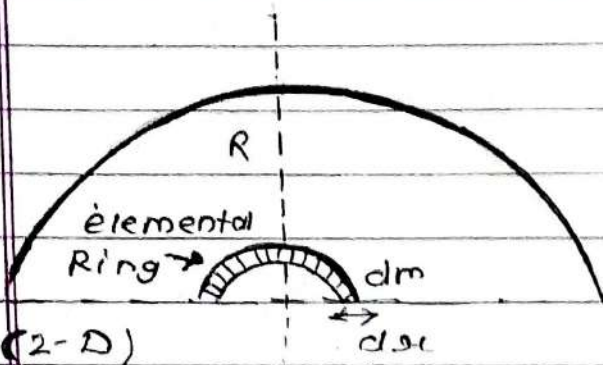
$$= R \lambda \times \pi \int_0^\pi \frac{-\cos \theta}{\pi R}$$

$$= \frac{R}{\pi} \left[-\cos \pi - (-\cos 0^\circ) \right]$$

$$= \frac{R}{\pi} \times 2$$

$$y_{\text{COM of ring}} = \frac{2R}{\pi}$$

3. COM of semicircular disc (Plate, Lamina) (2-D)



$$\text{Area} = dr \times \pi r$$

Mass per unit area = σ

$$\sigma = \frac{dm}{\text{Area}} = \frac{dm}{dr \times \pi r}$$

$$\Rightarrow dm = dr \pi r \sigma$$

$$\left[\sigma = \frac{M}{\pi R^2} = \frac{2M}{\pi R^2} \right]$$

$$y_{\text{com}} = \frac{\int y dm}{dm} = R \int_0^{2R} \frac{dr \pi r \sigma}{M} \quad \left[y \text{ of ring} = \frac{2R}{\pi} \right]$$

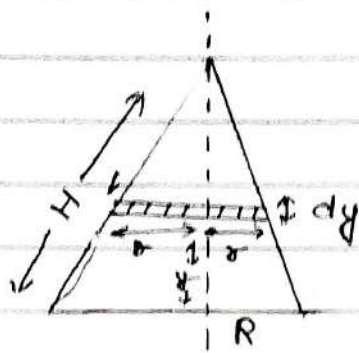
$$= \frac{2\sigma}{M} \left[r^2 dr \right]_0^R$$

$$= \frac{2 \times 2M}{M \times \pi R^2} \left[\frac{r^3}{3} \right]_0^R$$

$$= \frac{4}{\pi R^2} \times \frac{R^3}{3}$$

$$y_{\text{com}} \text{ of semi-circular disc} = \frac{4R}{3\pi}$$

4. Com of triangular lamina (isosceles) (2-D)



Both triangles are similar

$$\frac{r}{R} = \frac{H-y}{H}$$

$$r = \frac{R(H-y)}{H}$$

$$y_{com} = \frac{\int y dm}{M}$$

$$= \frac{\int y^2 dy \sigma}{M}$$

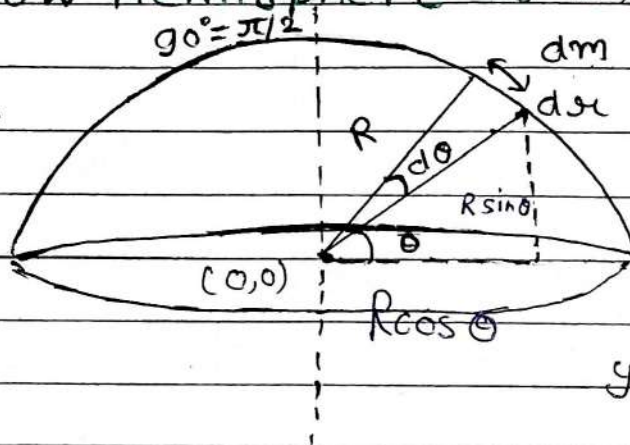
$$= \frac{\int y^2 \frac{(H-y)}{H} R \sigma dy}{M}$$

$$= \frac{2R\sigma}{HM} \int y (H-y) dy = \frac{2R \times M}{H \times M \times R H} \int_0^H (Hy - y^2) dy$$

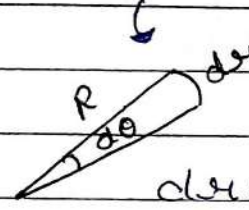
$$= \frac{2}{H^2} \left[\frac{Hy^2}{2} - \frac{y^3}{3} \right]_0^H = \frac{2}{H^2} \left[\frac{H^3}{2} - \frac{H^3}{3} \right] = \frac{2}{H^2} \left[\frac{H^3}{6} \right]$$

$$y_{com} \text{ of triangle from its base} = \frac{H}{3}$$

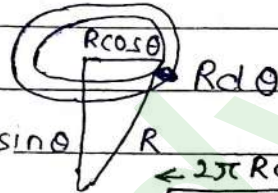
5. Hollow Hemisphere (3-D)



elemental ring



$$dA = R d\theta$$



Mass is distributed on surface = Area

$$\sigma = \frac{\text{Mass}}{\text{Area}} = \frac{M}{2\pi R^2}$$

$$dm = \sigma dA$$

$$= \sigma 2\pi R \cos \theta R d\theta$$

$$y_{\text{com}} = \frac{\int y dm}{\int dm} = \frac{\int_0^{\pi/2} R \sin \theta \sigma 2\pi R \cos \theta R d\theta}{\int_0^{\pi/2} dm}$$

$$= R^3 \pi \sigma \int_0^{\pi/2} 2 \sin \theta \cos \theta d\theta$$

$$= \frac{R^3 \times \pi \times M}{2\pi R^2} \int_0^{\pi/2} \sin 2\theta d\theta \quad (2 \sin \theta \cos \theta = \sin 2\theta)$$

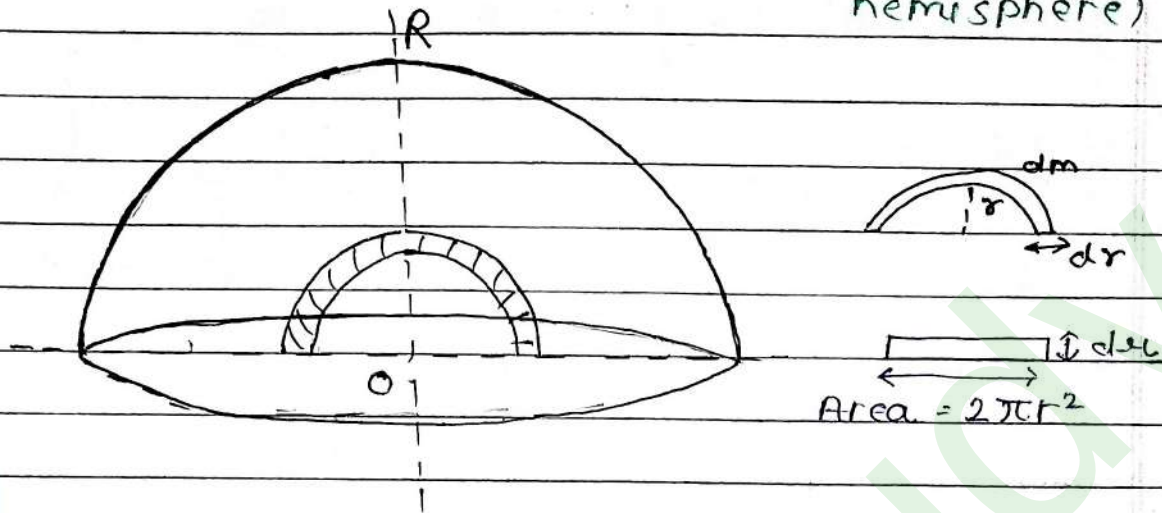
$$= \frac{R}{2} \left[\frac{-\cos 2\theta}{2} \right]_0^{\pi/2} \quad (\sin 2\theta = -\frac{\cos 2\theta}{2})$$

$$= \frac{R}{2} \left[\frac{-\cos 2\pi}{2} - \left(-\cos 2\theta^0 \right) \right]$$

$$= \frac{R}{2} \left[\frac{2}{2} \right]$$

$$y_{\text{com}} \text{ of a Hollow cylinder} = \frac{R}{2}$$

6. Solid Hemisphere (3-D) (∞ elemental hollow hemisphere)



3-D

$$\rho = \frac{M}{V} = \frac{M}{\frac{2}{3}\pi R^3} = \frac{3M}{2\pi R^3}$$

$$y_{com} = \int y \, dm$$

$$= \int_0^R \frac{y}{2} \rho 2\pi r^2 \, dh \quad \left[y = \frac{R}{2} \right]$$

$$dm = \rho \, dv$$

$$dm = \rho 2\pi r^2 \, dh$$

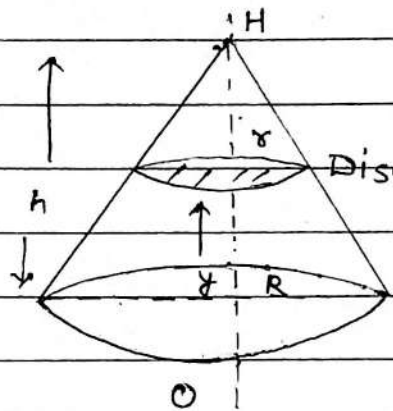
$$= \frac{\pi \rho}{M} \int_0^R r^3 \, dh$$

$$= \frac{\pi \times 3M}{M \times 2\pi R^3} \left[\frac{h^4}{4} \right]_0^R$$

$$= \frac{3}{2R^3} \times \frac{R^4}{4}$$

$$\text{COM of Solid hemisphere} = \frac{3R}{8}$$

7. Solid cone

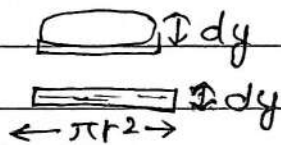


$$\frac{r}{R} = \frac{(H-y)}{H} \quad \left[r = \frac{R(H-y)}{H} \right]$$

$$\rho = \frac{M}{V} = \frac{M}{\frac{1}{3}\pi R^2 H} = \frac{3M}{\pi R^2 H}$$

$$dm = \rho dv$$

$$dm = \rho \pi r^2 dy$$



$$y_{com} = \frac{\int_0^H y dm}{dm} = \frac{\int_0^H y \rho \pi r^2 dy}{M}$$

$$= \frac{\rho \pi}{M} \int_0^H y [H-y]^2 R^2 dy$$

$$= \frac{3M \times \pi \times R^2}{M \times \pi R^2 H} \left[y (H-y)^2 dy \right]_0^H$$

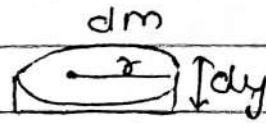
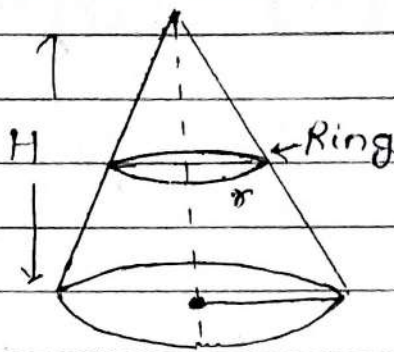
$$= \frac{3}{H^3} \left[\frac{H^2 y^2}{2} + \frac{y^4}{4} - \frac{2Hy^3}{3} \right]_0^H$$

$$= \frac{3}{H^3} \left[\frac{H^2 y^2}{2} + \frac{y^4}{4} - \frac{2Hy^3}{3} \right]_0^H$$

$$= \frac{3}{H^3} \times \frac{6H^4}{4} + \frac{3H^4}{4} - \frac{2H^4}{3} = \frac{1}{H^3} \times \frac{H^4}{4}$$

$$\text{COM of solid cone} = \frac{H}{4}$$

8 Hollow Cone



$$\frac{r}{R} = \frac{(H-y)}{H}$$

$$r = \frac{R(H-y)}{H}$$

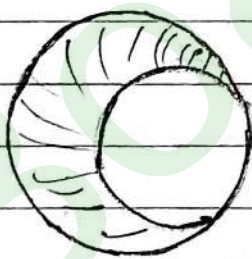
$$y_{com} = \frac{\int_0^H y dm}{M} = \frac{\int_0^H y \sigma \cdot 2\pi r dy}{M} = \frac{2\pi\sigma}{M} \int_0^H r y dy$$

$$= \frac{2 \times \pi \times M \times R}{M \times \pi \times R \times H \times L} \int_0^H (H-y) y dy = \frac{2}{H^2} \left[\frac{Ly^2}{2} - \frac{y^3}{3} \right]_0^H$$

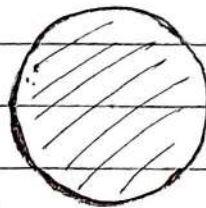
$$= \frac{2}{H^2} \times \frac{H^3}{3} = \frac{H}{3}$$

COM of Hollow cone = $\frac{H}{3}$

9. Centre of mass of Remaining Portion

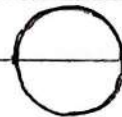


=

 M_1

(whole body)

+

 $-M_2$

(Removed part)

$$\vec{y}_{com} \text{ Remaining portion} = \frac{M_1 y_1 - M_2 y_2}{M_1 - M_2} = \frac{\text{Area}_1 y_1 - \text{Area}_2 y_2}{M_1 - M_2}$$

Motion of COM

$$x_{com} = \frac{m_1 x_1 + m_2 x_2 + \dots + m_n x_n}{m_1 + m_2 + \dots + m_n}$$

$$V_{com} = \frac{V_1 x_1 + V_2 x_2 + \dots + V_n x_n}{m_1 + m_2 + \dots + m_n}$$

$$a_{com} = \frac{m_1 a_1 + m_2 a_2 + \dots + m_n a_n}{m_1 + m_2 + \dots + m_n}$$

$$a_{com} (m_1 + m_2 + \dots + m_n) = \underbrace{m_1 a_1 + m_2 a_2 + \dots}_{F}$$

$$F_{net} = M a_{com}$$

$$F_{net} = 0$$

↓

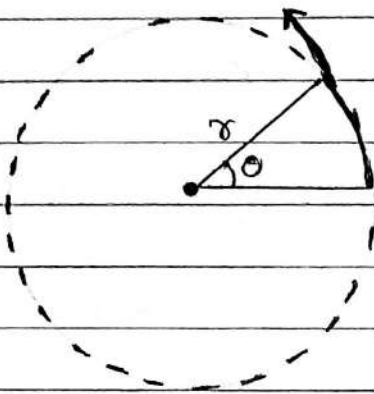
$$a_{com} = 0$$

↓

$$V_{com} = \text{Constant}$$

* IF $F_{net} = 0$ (system) and the body is initially at Rest, then system will remain at rest.

Dynamic of particles in circular motion.

LinearAngular1. Displacement (s) θ 2. Velocity (v) ω (omega)3. Acceleration (a_t/a_c) α 4. Force F τ (Torque)5. Mass m I (moment of Inertia)6. Momentum p L (Angular momentum)

$$s = r\theta$$

$$v_t = r\omega$$

$$v_{inst} = \frac{ds}{dt}, v_{ang} = \frac{\Delta s}{\Delta t}$$

$$\omega_{inst} = \frac{d\theta}{dt}, \omega_{ang} = \frac{\Delta \theta}{\Delta t}$$

* ω is same for all points of rotating body, but v is not due to radius.

$$[\omega_{pivoted point} = 0]$$

Ques IF $\omega = t^2 + 4$, find θ at $t = 2s$ if $\theta = 0$

Also calculate no. of revolution made in 2sec.

Soln

$$\omega = t^2 + 4$$

$$\omega = \frac{d\theta}{dt} \Rightarrow \int_{t_1}^{t_2} \omega dt = \int_{\theta_1}^{\theta_2} d\theta \Rightarrow \int_0^2 (t^2 + 4) dt = [\theta]_{\theta_1}^{\theta_2} = 0$$

$$\Rightarrow \theta_2 = \left[\frac{t^3}{3} \right]_0^2 + [4t]_0^2$$

$$2\pi = 1 \text{ revolution}$$

$$\Rightarrow \theta_2 = \frac{8}{3} + 8 = \frac{32}{3}$$

$$1 \text{ radian} = \frac{1}{2\pi} \text{ revol}$$

$$\theta \text{ radian} \times \frac{1}{2\pi} \times \theta = \frac{1}{2} \times \frac{7}{22} \times \frac{32}{3} = \frac{56}{33}$$

$\text{No. of revol.} = \frac{\theta}{2\pi}$
--

$$\begin{cases} \theta = \int \omega dt \\ s = \int v dt \end{cases}$$

+ $a_t, a_c, a_{net} \rightarrow$ Linear acceleration.

$\alpha \rightarrow$ angular acceleration (Rate of change of angular velocity)

$\alpha_{ang} = \frac{\Delta \omega}{\Delta t}$	$\alpha_{int.} = \frac{d\omega}{dt}$
---	--------------------------------------

$\alpha \rightarrow \text{rad/s}^2$

+ If v, R are constant then $|\omega| = \text{constant}$
 $\rightarrow \alpha, a_t = 0$

+ If ω & α have same sign $\rightarrow$ Body speeds up.

Equation of motion for uniform angular acceleration:-

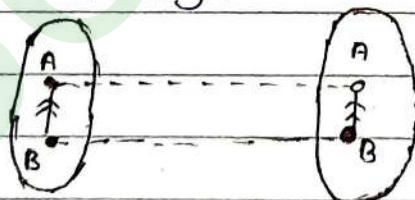
1. $\omega_f = \omega_i + \alpha t$

2. $\omega_f^2 = \omega_i^2 + 2\alpha\theta$

3. $\theta = \omega_i t + \frac{1}{2} \alpha t^2$

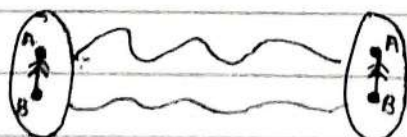
$\rightarrow$ Translation motion

- In pure translation motion, at any instant of time, all particles of the body have same velocity.



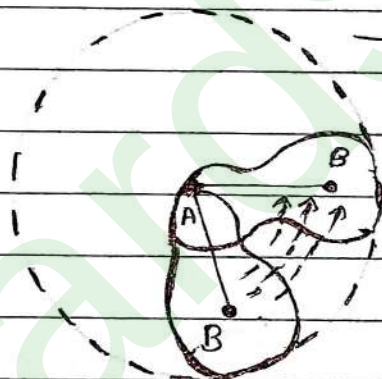
In this motion, lines of body joining A to B, After motion are parallel to each other.

- If parallel path is straight line $\rightarrow$ Rectilinear Translation motion.
- If parallel path is curve $\rightarrow$ curvilinear Translation motion



⇒ Rotational Motion

- In this motion, all particle moves in a circular path about a fixed axis. (axis of rotation)
- Line joining two internal points does not remains parallel to itself.



→ Not compulsory that is int.

Angular Momentum

→ It is defined as the moment of linear momentum of the particle about the axis.

→ Denoted by $\vec{L}$

→ Vector quantity, its magnitude is given by $L = r p \sin \theta$.

$$\boxed{\vec{L} = \vec{r} \times \vec{p}} \text{ always true}$$

→ Dimensions $\rightarrow ML^2 T^{-1}$

→ Unit $\rightarrow \text{kgm}^2/\text{s}$ or gcm^2/s

→ Angular momentum is find w.r.t an other point.

→ Direction of $\vec{L}$ is found by :-

(i) Right hand thumb rule

(ii) Screw Rule

(iii) $\vec{r}$ to $\vec{v}$, C.W $\rightarrow$ inward $\otimes$

A.C.W outward $\odot$

→ Special cases :-

(i) If $\theta = 0^\circ$ or 180° , $\sin \theta = 0$

$$\vec{L} = \vec{r} \times \vec{p} \Rightarrow L = r p \sin \theta = 0$$

→ Angular momentum is zero if line of action of p passes through the point of rotation.

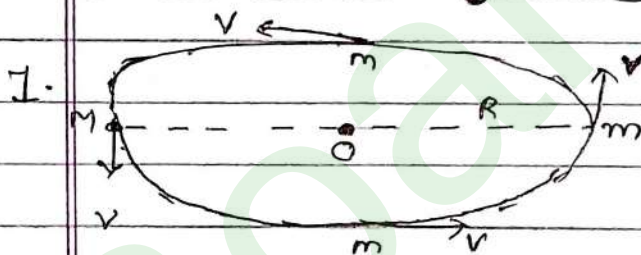
(ii) If $\theta = 90^\circ$, $\sin 90^\circ = 1$

→ $\vec{L}$ is maximum.

$$\begin{aligned} \vec{L} &= \vec{r} \times \vec{p} \\ &= \vec{r} \times (m\vec{v}) \\ &= m(\vec{r} \times \vec{v}) \\ &= mrv \sin \theta \\ &= mv(r \sin \theta) \end{aligned}$$

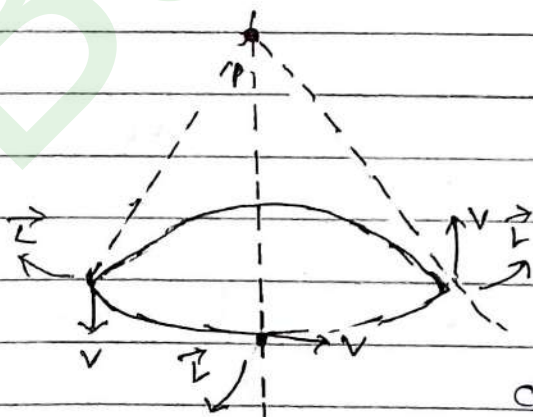
$$\boxed{\vec{L} = mvr \sin \theta \text{ or } mrv \sin \theta}$$

Cases of objects moving in a circle -



$$L_0 = mvr \sin \theta = mvr \quad (\uparrow)$$

L is constant in mag and direction.



$$L_p = mvr \sin \theta$$

→
- mag. constant but direction is variable
 $L_p = \text{variable}$.

$\swarrow$ τ
Torque (τ) [moment / turning effect of force]

+ Turning effect of force is called moment of force or torque.

+ It depends upon -

(i) magnitude of force $F(\uparrow) \propto (\uparrow)$

(ii) Perpendicular distance of line of action of force from axis of rotation.

$\swarrow$ $r(\uparrow) \propto (\uparrow)$

Moment or lever arm.

(iii) Angle at which F applied.

$F \sin \theta (\uparrow) \propto (\uparrow)$

+ Torque - Angular force.

+ Torque is the product of magnitude of the force and perpendicular distance between the line of action of force and axis of rotation.

$$\boxed{\text{Torque} = \text{force} \times \text{Lever arm}}$$

$$\boxed{\tau = \vec{F}_1 \times \vec{r}}$$

- + Dimension - $[ML^2T^{-2}]$ * Direction is determined by Right hand Thumb Rule ($\vec{R}$ to $\vec{F}$)
- + unit - Nm or dyne-cm
- + $\tau \rightarrow$ outward $\rightarrow$ Anti-clockwise $\Rightarrow +ve$
- + $\tau \rightarrow$ inward $\rightarrow$ clockwise $\Rightarrow -ve$

Note:- The component of τ along axis of rotation can only produce turning effect ($\theta = 0^\circ$ or 180°)

- + $\tau = 0$, when F is parallel to $\vec{r}$, F passes axis of rotation, F pass from pivoted point.
- + If $\theta = 90^\circ$, $\sin 90^\circ = 1 \rightarrow \tau = \text{maximum}$.

Q-2. A force $F = 2\hat{i} + 3\hat{j} - \hat{k}$ is acting at point $(1, 2, 2)^P$. Find the moment of force about the $(0, 1, 3)^O$

Solⁿ

$$\tau = \vec{F} \times \vec{r}$$

$$= (2\hat{i} + 3\hat{j} - \hat{k}) \times (\hat{i} + \hat{j} - \hat{k})$$

F applied

$$\vec{r} = \vec{r}_P - \vec{r}_O$$

$$= \hat{i}(-1 - (-3)) + \hat{j}(-1 - (-2)) + \hat{k}(3 - 2) = (\hat{i} + 2\hat{j} + \hat{k}) - (\hat{i} + \hat{j} + 3\hat{k})$$

$$= \hat{i} + 2\hat{j} - \hat{k}$$

i	j	k
1	1	-1
2	3	-1

$$\tau = 2\hat{i} - \hat{j} + \hat{k}$$

$$|\tau| = \sqrt{4 + 1 + 1} = \sqrt{6} \text{ Nm}$$

Moment of Inertia (I)

- Moment of Inertia of a rigid body about a fix axis is defined as the sum of product of masses of the particles constituting the body and the square of distances from axis of rotation.

$$I = m_1 r_1^2 + m_2 r_2^2 + \dots + m_n r_n^2$$

$$I = \sum_{i=1}^n m_i r_i^2$$

→ Dimension - ML^2

→ Unit - kgm^2

→ Moment of inertia of a body depends on:

- Mass of the body
- Size and shape of body
- Distributing of mass about axis of rotation.
- Position and orientation of axis of rotation w.r.t body.

→ MOI for continuous bodies, $I = \int dm r^2$

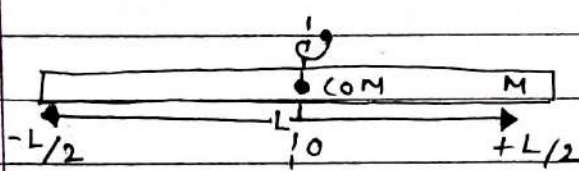
(Rotational)

Relation b/w I and KE :-

$$KE = \frac{1}{2} I \omega^2 = \frac{1}{2} (\sum m r^2) \omega^2$$

One Dimensional

1. moment of Inertia of a Thin uniform



Rod

1-D

 $\lambda = \text{mass per unit length}$

constant

1m Length $\rightarrow$ mass λ dx m length $\rightarrow$ mass $= dx \lambda$

$$I = \int_{-L/2}^{+L/2} dm x^2$$

$$dm = dx \lambda$$

$$I = \int_{-L/2}^{+L/2} dx \lambda x^2 = \int_{-L/2}^{+L/2} \lambda x^2 dx = \lambda \int_{-L/2}^{+L/2} x^2 dx$$

$$= \lambda \left[\frac{x^3}{3} \right]_{-L/2}^{+L/2}$$

$$\left[\lambda = \frac{M}{L} \right]$$

$$\text{Radius of Gyration (K)} \quad \lambda \left[\frac{L^3}{24} + \frac{L^3}{24} \right]$$

$$I = M K^2$$

$$= \frac{M}{L} \left[\frac{2L^3}{24} \right]$$

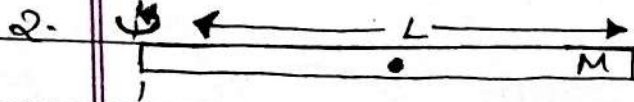
MOT of uniform rod along
the axis passing through

$$\text{com} - \left[I = \frac{ML^2}{12} \right]$$

$$MR^2 = \frac{ML^2}{12}$$

$$K = \sqrt{\frac{L^2}{12}} = \frac{L}{2\sqrt{3}}$$

Radius of
Gyration



$$I = \int_0^L dx \lambda x^2 = \int_0^L \lambda x^2 dx = \lambda \int_0^L x^2 dx$$

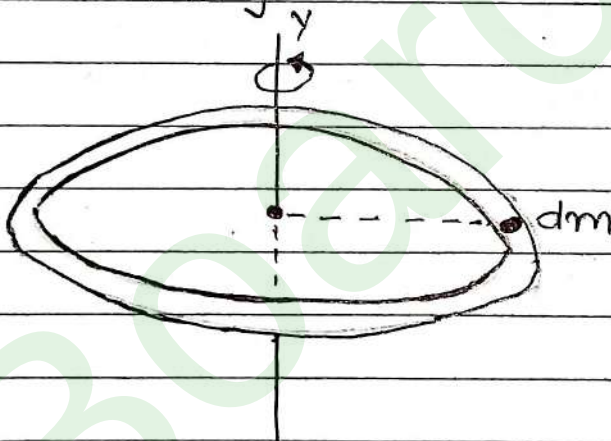
$$= \frac{M}{L} \left[\frac{x^3}{3} \right]_0^L$$

$$= \frac{M}{L} \times \frac{L^3}{3}$$

$$I = \frac{ML^2}{3}$$

Two-Dimensional

3. moI of a Ring (M, R) about an axis passes through COM and $\perp$ to plane of ring.



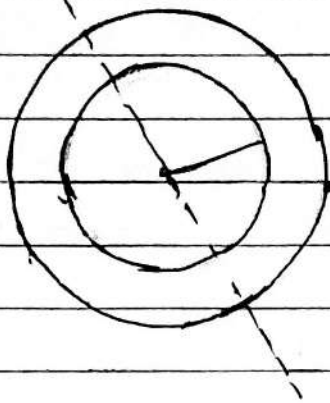
$$\int dI = \int dm R^2$$

$$I = R^2 M \quad (R = \text{constant})$$

$$I = MR^2$$

$$(dm + dm + \dots = M)$$

Q4

Mass is from R_1 to R_2 .moI of the disc about an axis passing through COM & $\perp$ to Plane of Axis.

$$\leftarrow 2\pi r \rightarrow$$

$$I = \int_{R_1}^{R_2} dm r^2$$

$$\int_{R_1}^{R_2} (2\pi r) dr \sigma r^2$$

$$\text{Area} = (2\pi r) dr$$

$$dm = (2\pi r) dr \sigma$$

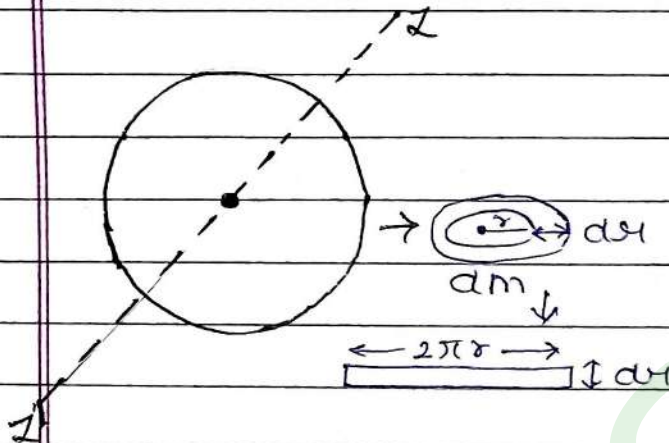
$$= 2\pi \sigma \left[\int_{R_1}^{R_2} r^3 dr \right]$$

$$= \frac{2\pi M}{\pi (R_2^2 - R_1^2)} \left[\frac{r^4}{4} \right]_{R_1}^{R_2} = \frac{2M}{R_2^2 - R_1^2} \left[\frac{R_2^4 - R_1^4}{4} \right]$$

$$\Rightarrow \boxed{I = \frac{M}{2} [R_1^2 + R_2^2]}$$

5. Disc

MoI about an axis passing through its COM and $\perp$ to plane of Disc (M, R).



2D

σ = mass per unit area.

1 m² area = σ mass

$$(2\pi r dr) dm = (2\pi r dr) \underbrace{\sigma}_{dm}$$

$$I = \int_0^R dm r^2$$

$$\left\{ \sigma = \frac{M}{\pi R^2} \right\}$$

$$I = \int_0^R (2\pi r dr) \sigma r^2$$

$$= 2\pi \sigma \int_0^R r^3 dr$$

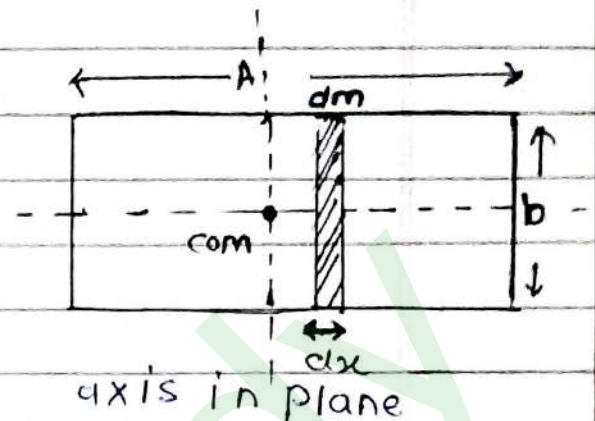
$$= \frac{2\pi M}{\pi R^2} \left[\frac{r^4}{4} \right]_0^R$$

$$= \frac{2M}{R^2} \left[\frac{R^4}{4} \right]$$

$$= \boxed{I = \frac{MR^2}{2}}$$

6. Rectangular Lamina

2D

 σ = Mass per unit area 1m^2 area = σ mass $b \times x$ area = $b \times x \times \sigma$ mass

$$\left[\sigma = \frac{M}{lb} \right]$$

$$I = \int_{-L/2}^{L/2} dm x^2$$

$$= \int_{-L/2}^{L/2} b dx \sigma x^2$$

$$= \sigma b \int_{-L/2}^{L/2} x^2 dx$$

$$= \frac{M}{lb} \times b \left[\frac{x^3}{3} \right]_{-L/2}^{L/2}$$

$$I = \frac{ML^2}{12}$$

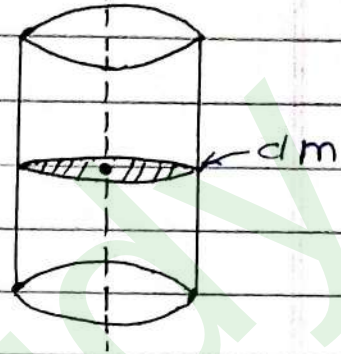
3-Dimensional

7. Hollow cylinder (Many elemental rings)

$$I = \int dm R^2$$

$R \rightarrow \text{constant}$

$$= R^2 M \quad \boxed{I = MR^2}$$



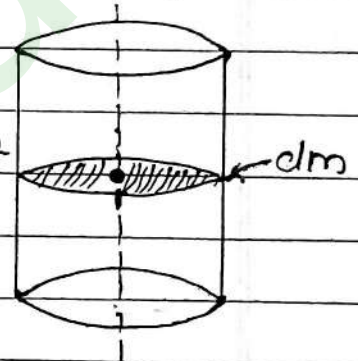
8. Solid cylinder (many elemental discs)

$$\int dI = \int dm \frac{R^2}{2}$$

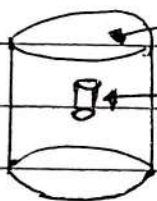
$$I = \frac{R^2}{2} \int dm$$

$$\boxed{I = \frac{MR^2}{2}}$$

disc $\rightarrow \frac{MR^2}{2}$



9.



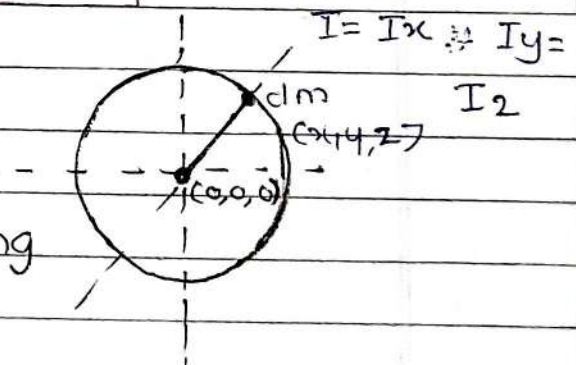
solid.

hollow

$$\boxed{I = \frac{M(R_1^2 + R_2^2)}{2}}$$

10. Hollow sphere

moI about any axis passing through COM



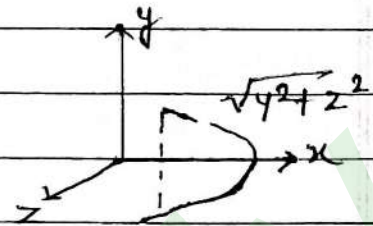
MOI of 'dm' mass about x axis

$$\text{Add } \begin{cases} dI_x = dm (y^2 + z^2) & (\text{dist. from x-axis})^2 \\ dI_y = dm (x^2 + z^2) \\ dI_z = dm (y^2 + x^2) \end{cases}$$

$$dm \, dI_x + dI_y + dI_z = 2dm(x^2 + y^2 + z^2)$$

$$dI_x = dI_y = dI_z = dI$$

$$\int 3dI = \int 2dm(x^2 + y^2 + z^2)$$



$$3I = \int 2dm R^2$$

$$3I = 2R^2 \int dm$$

$$3I = 2R^2 M$$

$$\text{dist. from } x \text{ axis} = \sqrt{y^2 + z^2}$$

$$\text{dist from } y \text{ axis} = \sqrt{x^2 + z^2}$$

$$\text{dist From } z \text{ axis} = \sqrt{x^2 + y^2}$$

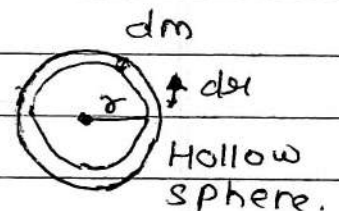
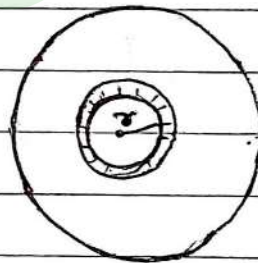
$$I = \frac{2MR^2}{3}$$

$$\vec{R} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$|\vec{R}| = \sqrt{x^2 + y^2 + z^2}$$

11 Solid Sphere (∞ hollow sphere)

MOT of Solid sphere
about any axis passing
through COM (M, R)



3-D

mass per unit Vol. = ρ

1 m³ vol. = ρ mass

$$4\pi r^2 dr \text{ m}^3 = (4\pi r^2 dr) \rho$$

$$\text{MOT } \frac{2}{3} M R^2$$

$$dI = \int_0^R \frac{2}{3} dm r^2$$

$$\int dI = \int_0^R \frac{2}{3} (4\pi r^2 dr) \rho r^2$$

$$= \frac{2}{3} 4\pi \rho \left[\frac{r^5}{5} \right]_0^R$$

$$= \frac{2}{3} \times 4\pi \times \frac{3M}{4\pi R^3} \left[\frac{r^5}{5} \right]_0^R$$



VOL of hollow sphere

$$\text{Vol} = \text{Area} \times L$$

$$= 4\pi r^2 \times dr$$

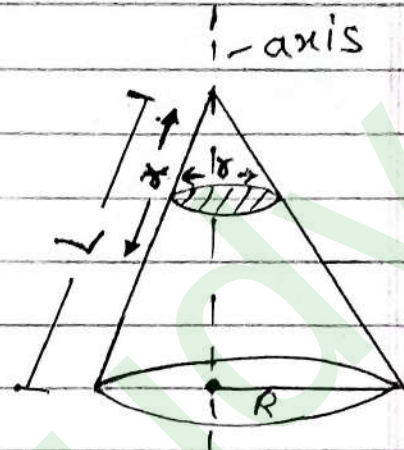
$$\rho = \frac{M}{\text{Vol}} = \frac{M}{\frac{4}{3}\pi r^3}$$

$$= \frac{3M}{4\pi r^3}$$

$$= \frac{2}{3} \times 4 \times \pi \times 3M \left[\frac{R^5}{5} \right]$$

$$I = \frac{2MR^2}{5}$$

12. **Hollow cone** (M, R, L)
(∞ elementary Ring)



Area of ring = $(2\pi r) dx$
 $\leftarrow 2\pi r \rightarrow$

$\rightarrow$ whole mass in one surface $\rightarrow$ area
mass unit area = σ

$$\left[\sigma = \frac{M}{\pi R L} \right]$$

CSA

$$(2\pi r dx) m^2 \text{ area} = (2\pi r) dx \sigma$$

$$I = \int dm x^2$$

$$= \int (2\pi r) dx \sigma x^2$$

Both triangle are similar
 $\frac{r}{R} = \frac{x}{L}$

$$r = \frac{Rx}{L}$$

$$= 2\pi \sigma \int_0^L x^3 dx$$

$$= 2\pi \sigma \int_0^L \frac{R^3 x^3}{L^3} dx$$

$$= 2\pi \times \frac{M}{\pi R L} \int_0^L \frac{R^3 x^3}{L^3} dx = \frac{2M \times R^3}{R L L^3} \left[\frac{x^4}{4} \right]_0^L$$

$$= \frac{2MR^2}{L^4} \times \frac{L^4}{4}$$

$$I = \frac{MR^2}{2}$$

(3-D)

13. Solid cone (M, R, h, L)
(∞ Element disc)

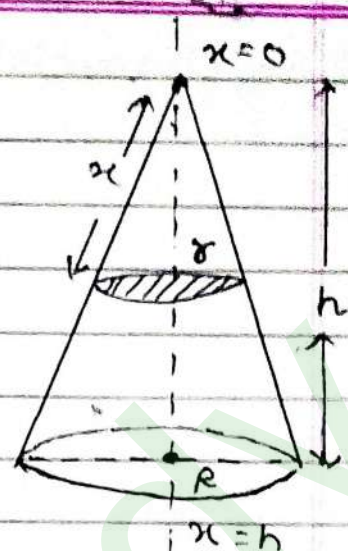


$$dm^3 vol = \rho$$

$$(\pi r^2 dx) m^3 \quad \boxed{\pi r^2} \quad dx$$

$$= (\pi r^2 dx) \rho$$

$\underbrace{\hspace{1cm}}_{dm}$



Both triangles are similar

$$\frac{r}{R} = \frac{x}{h}$$

$$r = \frac{Rx}{h}$$

$$\rho = \frac{M}{Vol} = \frac{M}{\frac{1}{3} \pi r^2 h}$$

$$I = \int dm \frac{y^2}{2}$$

$$I = \int \pi r^2 dx \rho \times \frac{y^2}{2}$$

$$= \frac{\pi}{2} \rho \int_0^h r^4 dx$$

$$= \frac{\pi}{2} \times \frac{3M}{\pi r^2 h} \int_0^h \frac{R^4 x^4}{h^4} dx = \frac{3M}{2R^2 h} \times \frac{R^4}{h^4} \left[\frac{x^5}{5} \right]_0^h$$

$$= \frac{3M}{2R^2 h} \times \frac{R^4 \times h^5}{h^4 \times 5}$$

$$= \frac{3MR^2}{10}$$

$$I = \frac{3MR^2}{10}$$

I of hollow cone > solid cone.

► Relation B/w Torque and Moment of Inertia

$$F = ma$$

$$\tau = rF$$

$$\tau_1 = r_1 m_1 a_1 \propto$$

$$\tau_1 = m_1 r_1^2 \alpha$$

$$\tau_2 = m_2 r_2^2 \alpha$$

$$\tau_{\text{net}} = m_1 r_1^2 \alpha + m_2 r_2^2 \alpha + \dots + m_n r_n^2 \alpha$$

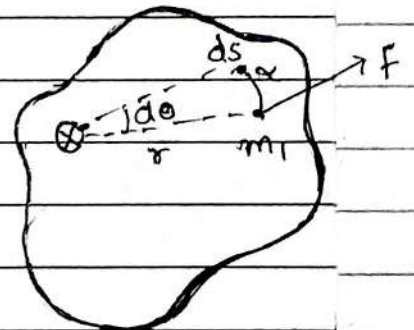
$$= \alpha [m_1 r_1^2 + m_2 r_2^2 + \dots + m_n r_n^2]$$

$$I = m_1 r_1^2 + m_2 r_2^2 + \dots$$

$$\tau_{\text{net}} = I \alpha$$

► Work done by torque
 w by torque = torque $\times$ angular disp.

$$\begin{aligned} dW &= F \times S \times \cos \theta \\ &= F ds \cos \alpha \\ &= (F \cos \alpha) ds \\ &= (F \cos \alpha) r d\theta \\ &= \underbrace{F r}_{\tau} d\theta \end{aligned}$$



$$\text{angle} = \frac{\text{arc}}{\text{Radius}}$$

$$\int dW = \int \tau d\theta$$

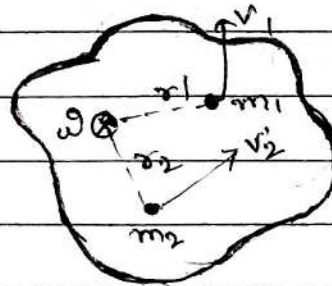
$$W = \int_{\theta_1}^{\theta_2} \tau d\theta$$

$$d\theta = \frac{ds}{r} \Rightarrow ds = r d\theta$$

Kinetic Energy of Rotating Bodies

$$KE_1 = \frac{1}{2} m_1 v_1^2$$

$$KE_2 = \frac{1}{2} m_2 v_2^2$$



$$K.E_n = \frac{1}{2} m_n^2 v_n^2$$

* Angular speed of every particle in a rotational motion is same.

$$K.E_{Total} = K.E_1 + K.E_2 + \dots + K.E_n$$

$$= \frac{1}{2} (m_1 v_1^2 + m_2 v_2^2 + \dots + m_n v_n^2)$$

$$= \frac{1}{2} (m_1 r_1^2 \omega^2 + m_2 r_2^2 \omega^2 + \dots + m_n r_n^2 \omega^2)$$

$$v_1 = r_1 \omega$$

$$v_2 = r_2 \omega$$

$$v_n = r_n \omega$$

$$= \frac{1}{2} \omega^2 (m_1 r_1^2 + m_2 r_2^2 + \dots + m_n r_n^2)$$

$$KE = \frac{1}{2} I \omega^2$$

* Work Energy Theorem can also be applied for (all forces = $\Delta K.E$) rotational motion.

$$\text{Proof } W = \int \tau d\theta$$

$$= \int I \alpha d\theta$$

$$= \int I \frac{d\omega}{dt} d\theta$$

$$= \int I d\omega \omega$$

$$= I \int_{\omega_i}^{\omega_f} \omega d\omega$$

$$= I \left[\frac{\omega^2}{2} \right]_{\omega_i}^{\omega_f}$$

$$= I \left[\frac{\omega_f^2}{2} - \frac{\omega_i^2}{2} \right]$$

$$= \frac{1}{2} I \omega_f^2 - \frac{1}{2} I \omega_i^2$$

$$W = \Delta K.E$$

Theorem proved

► Power delivered by Torque -

$$[\text{Power} = \text{Torque} \times \text{Angular velocity}]$$

Angular Momentum of purely Rotating Body

$$\vec{L} = \vec{r} \times \vec{p} = m v r \perp$$

$$L_1 = m_1 v_1 r_1 \text{ (outward)}$$

$$L_2 = m_2 v_2 r_2 \text{ (outward)}$$

!

$$L_n = m_n v_n r_n \text{ (outward)}$$

$$[L_{\text{net}} = L_1 + L_2 + \dots + L_n]$$

$$= m_1 v_1 r_1 + m_2 v_2 r_2 + \dots + m_n v_n r_n \quad \boxed{\vec{L} = I \vec{\omega}}$$

$$= m_1 r_1^2 \omega + m_2 r_2^2 \omega + \dots + m_n r_n^2 \omega$$

Valid for
Symmetrical bodies

$$= \omega = (m_1 r_1^2 + m_2 r_2^2 + \dots + m_n r_n^2)$$

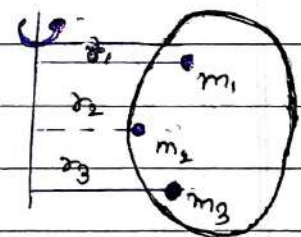
I

Relation b/w Angular momentum (L) and moment of Inertia

$$v_1 = r_1 \omega, \quad L_1 = m_1 r_1^2 \omega$$

$$v_2 = r_2 \omega$$

$$v_n = r_n \omega, \quad L_n = m_n r_n^2 \omega$$



$$L = L_1 + L_2 + \dots + L_n$$

$$= m_1 r_1^2 \omega + m_2 r_2^2 \omega + \dots + m_n r_n^2 \omega$$

$$= (m_1 r_1^2 + m_2 r_2^2 + \dots + m_n r_n^2) \omega$$

$$= (\sum m r^2) \omega$$

$$= I \text{ about a given axis}$$

$$\boxed{L = I \omega}$$

Angular momentum = I x Angular velocity

► Relation b/w torque and angular momentum:-

$$L = \vec{r} \times \vec{p}$$

(Differentiate in terms of 't')

$$\frac{dL}{dt} = \frac{d}{dt} (\vec{r} \times \vec{p})$$

Product rule

$$\frac{d(x+y)}{dt} = x \frac{dy}{dt} + y \frac{dx}{dt}$$

$$= \vec{r} \times \frac{d\vec{p}}{dt} + \vec{p} \times \frac{d\vec{r}}{dt} = \vec{r} \times \vec{F} + \vec{v} \times m\vec{v}$$

$$= \underbrace{\vec{r} \times \vec{F}}_{\tau} + \underbrace{m\vec{v} \times \vec{v}}_{=0}$$

$$\boxed{\frac{dL}{dt} = \tau}$$

Conservation of angular Momentum

Linear

$$\vec{F}_{\text{ext}} = \frac{d\vec{p}}{dt}$$

$$\vec{F}_{\text{ext}} = 0, \frac{d\vec{p}}{dt}$$

$$\vec{p} = \text{constant}$$

$$\vec{p}_i = \vec{p}_f$$

(system)

Rotational

$$\vec{\tau}_{\text{ext}} = \frac{d\vec{L}}{dt}$$

• Angular momentum will change in that direction as that of Torque (external)

$$\vec{\tau}_{\text{ext}} = 0, d\vec{L} = 0$$

$$\vec{L} = \text{constant}$$

$$\vec{L}_i = \vec{L}_f$$

If τ about an axis = 0, then L about that axis is constant.

Proof of $\vec{L}_i = \vec{L}_f$

$$\vec{L} = I\vec{\omega}$$

$$\frac{d\vec{L}}{dt} = I \frac{d\vec{\omega}}{dt}$$

If $\omega (\uparrow)$ then $I (\downarrow)$

$$\text{e.g. } \frac{\omega}{2}, 2I$$

$$\frac{d\vec{L}}{dt} = \frac{I\alpha}{\tau} \Rightarrow \tau = \frac{d\vec{L}}{dt} \quad \text{Theorem proved}$$